

MTF-Aware Deblurring with Physics-Guided Unrolled ADMM

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Abstract—Coded-exposure cameras redistribute motion blur energy to ease post-capture restoration, but reconstruction quality remains limited by realistic photon budgets, read noise, and the interaction of shutter codes with the optical transfer function (OTF/MTF). We present an MTF-aware deblurring framework that couples a physics-faithful forward simulator with classical, plug-and-play, and learned unrolled reconstruction pipelines. The forward model synthesizes box, random, and Modified Legendre Sequence (MLS) codes, computes PSF/OTF/MTF diagnostics, and injects Poisson–Gaussian noise. Baselines include Wiener filtering, damped Richardson–Lucy (RL), plug-and-play ADAM, and augmented-Lagrangian ADMM with strong denoisers (TinyCNN, DnCNN, UNet, DRUNet) plus MTF-aware scheduling. Building on this, we introduce a learnable unrolled ADMM variant that adapts per-iteration penalty, denoiser strength, and an optional learned MTF trust mask from scene/context features without retraining the denoiser. On DIV2K/X2 with blur length 15 px and photon budget 1000, classical RL averages ≈ 19 dB PSNR across codes, plug-and-play ADAM reaches ≈ 22 – 23 dB, and physics-aware ADMM with DRUNet peaks at 25–29 dB on a 10-image sweep. The unrolled model achieves 26.3/27.2/27.2 dB (box/random/legendre) on held-out data while matching or surpassing the hand-tuned physics-aware baseline on box/random and remaining competitive on legendre, with similar SSIM/LPIPS. The codebase runs on CPU, CUDA, or DirectML and ships with reproducible scripts, checkpoints, and evaluation logs to enable follow-on research in physics-guided deblurring.

Index Terms—Computational Photography

1 INTRODUCTION

MOTION blur from camera shake or object motion erases high-frequency content and introduces deep spectral nulls that make deblurring ill-conditioned, especially under limited light. Coded-exposure (flutter) shutters address this by modulating exposure within a frame so that spatial frequencies are redistributed rather than nulled, improving the conditioning of the inverse problem at capture time. Yet practical reconstruction remains challenging: photon budgets constrain SNR, read noise dominates dark regions, and code–optics interactions through the modulation transfer function (MTF) can negate theoretical gains. Classical deconvolution methods (Wiener, RL) either over-smooth or amplify noise under these conditions, while modern plug-and-play (PnP) methods often ignore the forward model’s structure when choosing denoiser schedules.

This work targets physics-aware deblurring under realistic noise and optical constraints. We make three contributions:

- 1) A reusable forward simulator that emits PSF/OTF/MTF diagnostics for box, random, and MLS codes with configurable blur length, duty cycle, photon budget, and read noise. It supports RGB/gray DIV2K crops and synthetic scenes and exports arrays/PNGs and spectral SNR plots for analysis.
- 2) A suite of baselines: Wiener filtering; damped and smoothed Richardson–Lucy; plug-and-play ADAM with TinyCNN/DnCNN/UNet priors; and physics-aware ADMM with DRUNet, including MTF



Fig. 1. Example one-column figure.

weighting, sigma adaptation, and a scheduler that ties the augmented-Lagrangian penalty to the measured degradation.

- 3) A learnable unrolled ADMM variant that adapts per-iteration parameters—penalty (ρ), denoiser weight, sigma multiplier—and an optional learned MTF trust mask via a lightweight context MLP, while keeping the denoiser frozen. This reduces hand-tuning and captures scene-dependent degradation cues.

Empirically, physics-aware ADMM with DRUNet attains 25–29 dB PSNR on a 10-image DIV2K/X2 sweep, while the unrolled model improves box/random codes on held-out data and remains competitive on legendre. The framework is device-agnostic (CPU, CUDA, DirectML) and ships with CLI entry points, dataset loaders, and checkpoints to enable replication and extension.

2 RELATED WORK

Coded exposure and flutter shutters. Early flutter shutters (e.g., random or MLS/Legendre codes) modulate exposure to avoid deep frequency nulls and yield invertible motion PSFs under small-motion assumptions. Our simulator reproduces these codes and surfaces their frequency responses

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through PSF/OTF/MTF plots, exposing how duty cycle, taps, and blur length shape conditioning.

Classical deconvolution. Wiener filtering and Richardson–Lucy (RL) remain standard baselines but are sensitive to Poisson–Gaussian noise. We observe RL at ≈ 19 dB PSNR on full DIV2K/X2 (800 images), highlighting the need for stronger priors when photon budgets are constrained.

Plug-and-play optimization. PnP methods inject denoisers into iterative solvers (ADMM, HQS, gradient methods). Our plug-and-play ADAM baseline with TinyCNN/DnCNN/UNet priors reaches ≈ 22 –23 dB PSNR on DIV2K/X2, outperforming RL by 3–4 dB. However, fixed schedules disregard the forward model’s conditioning.

Physics-aware ADMM. Augmented-Lagrangian ADMM with strong denoisers (e.g., DRUNet) is robust and extensible. We add MTF weighting, sigma adaptation, and a scheduler tied to the degradation, improving RGB performance to 25–29 dB on a 10-image sweep ($\rho \approx 1.04$, denoiser weight ≈ 0.16 , scheduler enabled).

Learned unrolling. Unrolled HQS/ADMM learns step sizes and proximal strengths. Our variant conditions per-iteration parameters and an MTF trust mask on context features (blur length, taps, photon budget, optics/SNR scores) while freezing the denoiser, balancing adaptability with stability under a consistent noise model.

3 PROPOSED METHOD

3.1 Forward Model

We model coded exposure as a linear convolution with additive noise:

$$y = k * x + n \quad (1)$$

The simulator:

- Generates exposure codes (box, random, MLS) with taps T , blur length L , and duty cycle.
- Converts codes to PSFs via a motion kernel and computes OTF/MTF diagnostics.
- Injects Poisson–Gaussian noise parameterized by photon budget and read-noise σ .
- Supports RGB or grayscale DIV2K crops and synthetic scenes; exports arrays, PNGs, and spectra.

3.2 Classical Baselines

- **Wiener:** frequency-domain deconvolution with tunable k (noise-to-signal ratio). Fast but prone to ringing at high noise.
- **Richardson–Lucy:** iterative EM-style deconvolution with damping, TV, and Gaussian smoothing to control noise amplification. We use 12 iterations, damping 0.7, TV $1e-3$, smoothing weight 0.4, $\sigma = 1.5$.

3.3 Plug-and-Play ADAM

We optimize the data term with ADAM and inject a denoiser every m steps. Key hyperparameters: learning rate, denoiser weight, interval m . Priors are swappable (TinyCNN, DnCNN, UNet). The Tiny prior ($\sigma \approx 15/255$) is CPU-friendly; DnCNN and UNet benefit from GPU acceleration.

3.4 Physics-Aware ADMM

We use FFT-domain x-updates and a denoiser proximal. Physics hooks:

- **MTF weighting:** adjusts the data fidelity step using MTF-derived weights (modes: none, gamma, wiener, combined).

$$\min_x \frac{1}{2} \|W \cdot (k * x - y)\|^2 + \lambda \mathcal{R}(x) \quad (2)$$

- **Sigma adaptation:** adjusts denoiser strength based on MTF and noise settings.
- **Scheduler:** varies ρ and denoiser weight over iterations conditioned on degradation parameters.

DRUNet priors yield the strongest RGB performance.

3.5 Learnable Unrolled ADMM

We unroll K iterations with trainable per-iteration $\{\rho, \text{denoiser weight, sigma multiplier}\}$ and an optional learned MTF trust mask (gamma, floor, cutoff). A lightweight context MLP consumes normalized features [blur_length, taps, photon, optics, snr] and produces deltas applied to base parameters. The denoiser is frozen (UNet backbone); only unrolled parameters and MTF mask are learned. Training is end-to-end on DIV2K with on-the-fly forward simulation, minimizing reconstruction loss under the same photon/read noise used at inference.

4 EXPERIMENTAL RESULTS

5 EXPERIMENTAL SETUP

- **Data:** DIV2K_train_LR_bicubic/X2. For RGB sweeps we use 256×256 crops; grayscale smokes use 128×128 . Synthetic scenes are used for quick sanity checks.
- **Forward settings:** blur length 15 px, taps $T = 31$, duty cycle 0.5, photon budget 1000, read noise sigma 0.01. Codes: box, random, MLS (legendre).
- **Metrics:** PSNR, SSIM, LPIPS. Qualitative crops and spectral SNR plots for visual analysis.
- **Hardware:** CPU-only runs are supported; GPU runs target CUDA or DirectML. Training of the unrolled model used an RTX 4090; inference works on CPU/CUDA/DirectML via CLI flags.
- **Reproducibility:** CLI entry points under `mtf_aware_deblurring.pipelines.reconstruct` and `scripts/train_unrolled.py`; checkpoints stored in `checkpoints/`; logs and CSVs in `forward_model_outputs/` and `logs/`.

6 RESULTS

6.1 Classical and PnP Baselines

- **Richardson–Lucy:** ≈ 19.0 dB PSNR across box/random/legendre on 800 RGB images (DIV2K/X2, 256×256 crops). RL benefits from damping and smoothing but remains noise-limited.
- **Plug-and-play ADAM (Tiny prior):** 22.16/22.78/22.69 dB (box/random/legendre), a +3–4 dB gain over RL. DnCNN improves small grayscale smokes to ≈ 23.8 dB on legendre; UNet is competitive when GPU-backed.



Fig. 2. Example two-column figure.

6.2 Physics-Aware ADMM

- **RGB 10-image sweep (DRUNet, scheduler on):** $\rho \approx 1.04$, denoiser weight ≈ 0.16 , 60 iters, MTF weighting combined, sigma adaptation enabled $\rightarrow$ 25.34 dB (box), 28.43 dB (random), 29.38 dB (legendre). Nearby ρ /weight pairs (1.00–1.02, 0.15–0.16) are effectively tied.

6.3 Learnable Unrolled ADMM

- **Single-image eval (checkpoint checkpoints/full_run_21482/unrolled_best.pt via CLI):** 26.34 dB / 0.703 SSIM / 0.381 LPIPS (box), 27.19 dB / 0.789 / 0.293 (random), 27.15 dB / 0.787 / 0.310 (legendre). Matches or surpasses tuned physics-aware ADMM on box/random; slightly trails on legendre but remains within ≈ 1 dB.
- **Training behavior:** Stable with $K \approx 24$ steps, batch=2, denoise every 4, learning rate $1e-4$, MTF mask enabled. Higher steps (31) OOMed; LR warmups or cosine schedules underperformed flat LR.

6.4 Qualitative Observations

- Coded shutters preserve high-frequency structure versus box; MLS reduces ringing relative to random codes.
- Physics-aware ADMM reduces halo artifacts compared to RL/ADAM, especially near edges aligned with motion.
- Unrolled ADMM produces cleaner backgrounds on box/random where the scheduler alone struggled, suggesting the learned MTF trust mask helps suppress overconfident updates in low-SNR bands.

7 DISCUSSION AND LIMITATIONS

- **Noise-model fidelity:** Results assume Poisson–Gaussian noise with fixed photon budget and read noise; distribution shifts (different sensors or exposure settings) may require denoiser adaptation or re-training the unrolled parameters.
- **Denoiser dependence:** Performance hinges on the frozen prior. Stronger priors (e.g., diffusion) could further improve results but would increase compute and memory.
- **Real-data gap:** All results are simulated; real captures may exhibit spatially varying blur, rolling shutter, or lens aberrations not modeled here. Incorporating measured PSFs/MTFs is a natural extension.

TABLE 1
An Example of a Table

One	Two	One	Two
Three	Four	Three	Four
Three	Four	Three	Four
Three	Four	Three	Four

- **Compute constraints:** Unrolled models with large K can OOM on GPUs with limited memory; mixed precision or checkpointing could mitigate this.

8 FUTURE WORK

- Integrate learned or optimized shutter codes within the same framework to co-design capture and reconstruction.
- Add validation splits and early stopping for unrolled training; explore mild LR drops post-plateau.
- Benchmark diffusion-based priors inside the physics-aware ADMM and unrolled variants.
- Collect a small real dataset with measured PSFs/MTFs to quantify domain gap and refine the forward model.
- Explore robustness to heavier motion, lower photon budgets, and nonuniform blur fields.

9 CONCLUSION

We presented an MTF-aware deblurring toolkit that unifies physics-faithful simulation, classical baselines, plug-and-play optimization, and a learnable unrolled ADMM tailored to coded-exposure imaging. By exposing MTF/OTF diagnostics and conditioning reconstruction schedules on the observed degradation, the framework reduces reliance on hand-tuning and improves reconstruction quality under realistic noise. The released code, checkpoints, and scripts provide a reproducible foundation for future work on physics-guided deblurring and joint capture–reconstruction design.

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Michael Shell Biography text here.

John Doe Biography text here.